N O R M A T I V E

`T = I    det = +1    (x, y) -> (x, y)`

"This is the standard."

RFC 2119 MUST / SHALL

Owl Semaphore

System Specification — A Finite Algebra of Epistemic States

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SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v1.3.0-rc · LICENSE CC-BY-4.0

BANNER-TUPLE :: STATE=SYSTEM :: TRANSFORM=T = I   det = +1   (x, y) -> (x, y) :: QUOTE="This is the standard." :: VERSION=v1.3.0-rc :: CONCEPT-DOI=10.5281/zenodo.19473697 :: PUBLISHED-VERSION-DOI=10.5281/zenodo.19474599 :: RC-VERSION-DOI=TBD_BY_ZENODO_ON_RELEASE

OWL SEMAPHORE SYSTEM — MASTER PROOF

A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.

1. Statement of Intent

This document defines the Owl Semaphore as a formal epistemic system grounded in mathematics, perception, and reproducible graphical structure.

This is not a taxonomy of labels. It is a closed algebra over epistemic states, mapped into a constrained visual notation system.

The objective is a system that is:

- mathematically coherent
- visually invariant
- epistemically meaningful
- operationally reproducible
- resistant to ambiguity and drift

1.1 Canonical Sentence Stack (v1.3.0-rc)

The project uses a three-layer canonical sentence stack so that the same concept can be expressed at the level of mathematics, operation, and human story without drifting between documents:

Layer	Canonical sentence	Use
Formal	*A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.*	README, system spec, Zenodo metadata draft, citation abstract
Operational	*A four-state visual system for marking how a claim, document, dataset, or finding should be evaluated before belief, challenge, or action.*	Explanation document, public overview, DNS Tool bridge
Human	*Four owls tell the reader what kind of thinking they are looking at: standard, exploration, inversion, or self-audit.*	Story sections, teaching material

Earlier inconsistent forms — *"mapped into a visual system with strict invariants"* (former §11) and *"implemented as a reproducible visual system with enforced invariants"* (former README masthead) — are reconciled in v1.3.0-rc onto the canonical formal sentence above.

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2. Mathematical Foundation

2.1 Group Definition

The Owl Semaphore's discrete state space is modeled as the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

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This is a finite subgroup of the orthogonal group $O(2)$ isomorphic to V_4 (equivalently the dihedral group D_2); it is not O(2) itself (Klein four-group, Wikipedia; nLab; Knill, Harvard Math 22b, Unit 8: O(2)).

2.2 Elements

State	Operator	Mapping	Determinant
NORMATIVE	I	(x,y) → (x,y)	+1
NON-NORMATIVE	σ_v	(x,y) → (-x,y)	−1
CRITICAL	C_2	(x,y) → (-x,-y)	+1
METACOGNITIVE	σ_h	(x,y) → (x,-y)	−1

The σ_h assignment to METACOGNITIVE is unchanged from v1.2.0. Only the interpretive wording is refined in v1.3.0-rc (see §4).

2.3 Closure (Cayley Table)

The system is closed under composition. The Cayley table is:

$\circ$	I	σ_v	C_2	σ_h
I	I	σ_v	C_2	σ_h
$\boldsymbol{\sigma_v}$	σ_v	I	σ_h	C_2
$\mathbf{C_2}$	C_2	σ_h	I	σ_v
$\boldsymbol{\sigma_h}$	σ_h	C_2	σ_v	I

Each element is its own inverse: $g^2 = I$ for all $g \in V_4$. The four group axioms (closure, associativity, identity, inverses) hold by the table above.

2.4 Interpretation

This closure is not decorative. It enforces that all epistemic transitions remain inside a defined state space. Group structure guarantees algebraic closure; it does not by itself guarantee security or behavioral correctness without further proof.

3. State vs Process

3.1 Discrete States

The four owls represent discrete epistemic states:

- identity (NORMATIVE)

- reflection (NON-NORMATIVE)
- inversion (CRITICAL)
- frame-audit (METACOGNITIVE)

3.2 Continuous Process

Not all operations belong to the state system.

3.3 The 31° Rotation

The measured ~31° rotation is not part of V_4.

- it is not closed
- repeated composition does not return to the set

It represents **process**, not state.

3.4 Principle

States classify position. Processes move between positions.

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4. Epistemic Model

4.1 Core Structure

The system separates three levels:

1. object of analysis
2. observer
3. evaluative frame

4.2 State Mapping (Normative Phrasing — v1.3.0-rc)

State	Quote (scientific / normative)	Meaning
NORMATIVE	*"This is the standard."*	baseline framework
NON-NORMATIVE	*"This reflects the standard."*	reflected interpretation
CRITICAL	*"This inverts the standard."*	inverted assumptions
METACOGNITIVE	*"The observer audits the frame."*	observer audits its own evaluative frame — thinking about thinking

> **Note on the METACOGNITIVE phrasing.** The earlier line *"This audits the standard"* is deprecated as of v1.3.0-rc. The audit at METACOGNITIVE is directed at the **observer's**

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own evaluative frame, not at the standard as an external object. The explanatory variant — *"Thinking examines its own frame"* — appears in the warmer-voiced OWL-SEMAPHORE-EXPLANATION.md. This refinement aligns the language with the cognitive-science meaning of metacognition: monitoring and regulation of one's own cognitive process (metacognitive reflection review, PMC 11368986).

4.3 Critical Distinction

The system is only valid if these states are not conflated.

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5. Physical Grounding

The system is grounded in observable behavior.

5.1 Canonical Example

The METACOGNITIVE state is physically instantiated by:

- searching a space normally
- failing to detect a target
- inverting the viewing frame

5.2 Interpretation

The object does not change. The observer does not change as an agent. The frame changes.

In short: the observer audits the frame.

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6. Visual System Mapping

6.1 Shared Geometry

All owls share:

- identical center
- identical radial structure
- identical meander ring

6.2 Transform Consistency

Each owl is derived from the normative form by a valid element of V_4.

No arbitrary transformations are permitted.

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6.3 Invariants

- geometry is fixed
- center is fixed
- ring structure is fixed

7. Color System and Accessibility

Each state is assigned a distinct color space region:

- gold → normative authority
- teal → analytical reflection
- red → adversarial inversion
- amethyst → introspective frame-audit

7.1 Constraint

Color is semantic, not decorative.

7.2 Accessibility — Triple-Redundant Encoding (v1.3.0-rc, normative)

Color is not the only carrier. State identity must remain recoverable when color is removed (grayscale rendering, color vision deficiency, or low-vision contexts). Every state in the Owl Semaphore is therefore encoded through at least three independent channels:

1. **color** (the palette assigned in §7)
2. **orientation** (the V_4 transform applied to the canonical owl: upright/inverted × right/left-facing)
3. **textual label and context** (the literal state token — `NORMATIVE`, `NON-NORMATIVE`, `CRITICAL`, `METACOGNITIVE` — and the math/quote tuple printed alongside the badge)

This satisfies the design intent of **WCAG 2.2 SC 1.4.1 (Use of Color, Level A)**, which prohibits color from being the only visual means of conveying information (W3C), and aligns with Section 508 §302.3 (Section 508.gov). The CRITICAL state's intentionally low red-on-red contrast is the most acute test of this rule: redness alone never carries CRITICAL identity — orientation (upside-down, left-facing) and the literal label `CRITICAL` are required.

Red-green color vision deficiency affects approximately 8% of males and 0.5% of females of Northern European descent; rates vary by population (PMC global CVD review, 12385717).

Full WCAG 2.2 Level AA empirical conformance testing (automated checks, CVD simulation, user testing) is scoped to a future release; v1.3.0-rc states the design rule and its compliance intent.

8. Integrity Model

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All assets must satisfy:

- reproducibility from layers
- RGBA transparency correctness
- cryptographic verification (SHA-3-512)

8.1 Principle

An asset is not valid because it looks correct. It is valid because it verifies.

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9. File and System Architecture

9.1 Structure

OWL-SEMAPHORE/ ├── OWL-SEMAPHORE-SYSTEM.md ├── OWL-SEMAPHORE-EXPLANATION.md ├── INTEGRITY-MANIFEST.md ├── CHANGELOG.md ├── OWL-1-NORMATIVE.md ├── OWL-2-NON-NORMATIVE.md ├── OWL-3-CRITICAL.md └── OWL-4-METACOGNITIVE.md

9.2 Separation of Concerns

- system rules live in the system spec
- state rules live in the four owl-specific files
- origin story and audience rationale live in the explanation document
- canonical sentence history lives in `CHANGELOG.md`

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10. Interpretation Doctrine

10.1 The System Encodes Position, Not Truth

The Owl Semaphore does not assert correctness.

It encodes:

- how something is being evaluated
- not whether it is ultimately true

10.2 Misuse Condition

If the system is used to imply certainty rather than epistemic position, it is being used incorrectly.

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11. Core Principle (Reconciled, v1.3.0-rc)

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This system is defined as:

> **A finite algebra over epistemic states, implemented as a reproducible visual notation system with enforced invariants.**

This sentence is the single formal canonical definition for v1.3.0-rc. It supersedes both *"implemented as a reproducible visual system with enforced invariants"* and *"mapped into a visual system with strict invariants"*. See CHANGELOG.md for the per-version canonical-sentence history.

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12. Normative-Language Discipline

Where the Owl Semaphore documents use the RFC 2119 / RFC 8174 keywords (MUST, MUST NOT, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, NOT RECOMMENDED, MAY, OPTIONAL), they carry their BCP 14 meaning **only when they appear in all capitals** (RFC 2119; RFC 8174). Lowercase forms ("must", "should", "may") carry ordinary English meanings.

Sections labeled "(normative)" form part of the canonical specification; sections labeled "(informative)" or "(explanatory)" provide context only and do not extend the algebra.

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13. Closing Statement

The Owl Semaphore is designed to make reasoning visible.

It enforces structure where ambiguity would otherwise dominate, and it preserves interpretability across transformation, disagreement, and self-examination.

It is not decoration.

It is a constraint system for thought.

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OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v1.3.0-rc)

NORMATIVE

$T = I \quad \det = +1$

(x, y) -> (x, y)

"This is the standard."

RFC 2119 MUST / SHALL

NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

(x, y) -> (-x, y)

"This reflects the standard."

Informative / Advisory (NOTE)

CRITICAL

$T = C_2 \quad \det = +1$

(x, y) -> (-x, -y)

"This inverts the standard."

RFC 2119 MUST NOT / SHALL NOT

METACOGNITIVE

$T = \sigma_h \quad \det = -1$

(x, y) -> (x, -y)

"The observer audits the frame."

Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).

Owl Semaphore v1.3.0-rc · github.com/IT-Help-San-Diego/owl-semaphore
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